

Name: Phan Tran Thanh Huy

ID: ITCSIU22056

Lab 1

I1

Problem 1.1

```
1  #include <stdio.h>
2  int main() {
3      printf("Welcome to Geeks for Geeks!\n");
4      return 0;
5  }
6
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
root@DESKTOP-J52EF2R:~/projects/cpp# gcc welcome.c
root@DESKTOP-J52EF2R:~/projects/cpp# ./welcome.c
-bash: ./welcome.c: Permission denied
root@DESKTOP-J52EF2R:~/projects/cpp# a.out
a.out: command not found
root@DESKTOP-J52EF2R:~/projects/cpp# ./a.out
Welcome to Geeks for Geeks!
root@DESKTOP-J52EF2R:~/projects/cpp#
```

Problem 1.2

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      cout << "Welcome To Geeks For Geeks For Geeks";
5      return 0;
6  }
```

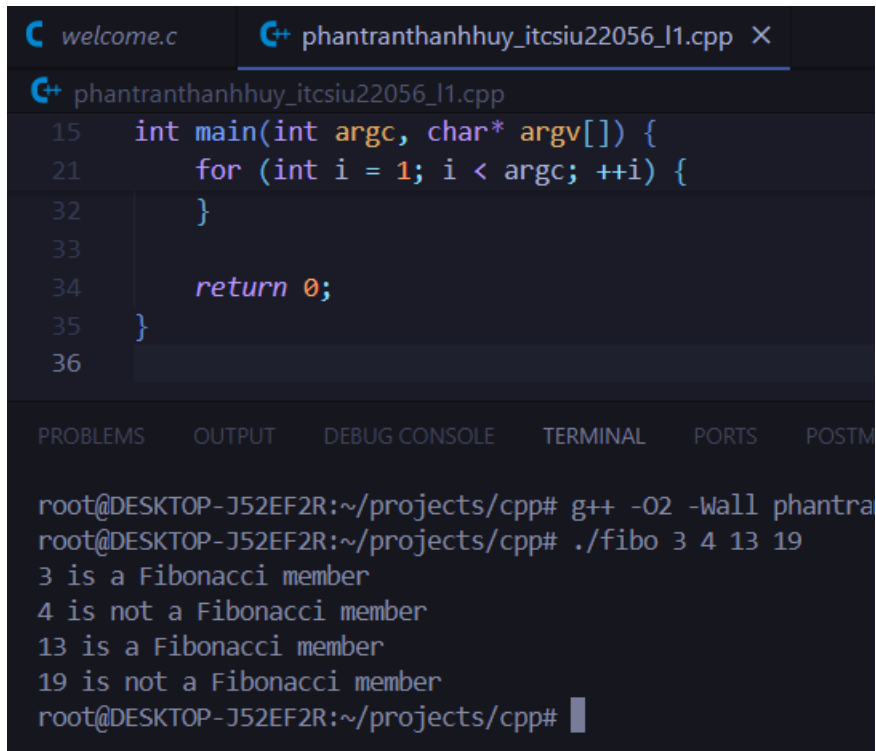
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
root@DESKTOP-J52EF2R:~/projects/cpp# g++ cplusplus.cpp
root@DESKTOP-J52EF2R:~/projects/cpp# ./a.out
Welcome To Geeks For Geeks For Geeksroot@DESKTOP-J52EF2R:~/projects/cpp#
```

I21

Problem 1.1

Problem 1.2



The screenshot shows a C++ IDE with two tabs: `welcome.c` and `phantranthanh Huy_itcsiu22056_l1.cpp`. The active tab displays the following C++ code:

```
15 int main(int argc, char* argv[]) {  
21     for (int i = 1; i < argc; ++i) {  
32     }  
33  
34     return 0;  
35 }  
36
```

Below the code editor, the `TERMINAL` tab is active, showing the following commands and output:

```
root@DESKTOP-J52EF2R:~/projects/cpp# g++ -O2 -Wall phantra  
root@DESKTOP-J52EF2R:~/projects/cpp# ./fibo 3 4 13 19  
3 is a Fibonacci member  
4 is not a Fibonacci member  
13 is a Fibonacci member  
19 is not a Fibonacci member  
root@DESKTOP-J52EF2R:~/projects/cpp#
```

I22

Problem 1.1

Problem 1.2

```
1  #include <iostream>
2  #include <cmath>
3  #include <cstdlib>
4  using namespace std;
5
6  bool isPrime(long long x) {
7      for(long long i=2; i<=sqrt(x); i++) {
8          if(x % i == 0) return false;
9      }
10     return true;
11 }
12
13 int main(int argc, char* argv[]) {
14     if (argc < 2) {
15         cout << "Usage: " << argv[0] << " <num1> <num2> <num3> ..." << endl;
16         return 1;
17     }
18
19     for (int i = 1; i < argc; ++i) {
20         long long n = atoll(argv[i]);
21         if (n <= 0) {
22             cout << "Error: " << argv[i] << " is not a positive integer." << endl;
23             continue;
24         }
25
26         if (isPrime(n))
27             cout << n << " is a prime" << endl;
28         else
29             cout << n << " is not a prime" << endl;
30     }
31
32     return 0;
33 }
34
```

```
root@DESKTOP-J52EF2R:~/projects/cpp# g++ -O2 -Wall phantranthanhuy_itcsiu22056_l22.cpp -o fibo
root@DESKTOP-J52EF2R:~/projects/cpp# ./fibo 12 3 19 4
12 is not a prime
3 is a prime
19 is a prime
4 is not a prime
root@DESKTOP-J52EF2R:~/projects/cpp#
```

I23

Problem 1.1

Problem 1.2

```
1
2 #include <iostream>
3 #include <cmath>
4 #include <cstdlib>
5 using namespace std;
6
7 bool isGrowing(long long x) {
8     if (x < 10) return false;
9     int n = 10;
10    while(x > 0) {
11        int last_digit = x % 10;
12        x /= 10;
13        if (last_digit >= n)
14            return false;
15        else {
16            n = last_digit;
17        }
18    }
19    return true;
20 }
21
22 int main(int argc, char* argv[]) {
23     if (argc < 2) {
24         cout << "Usage: " << argv[0] << " <num1> <num2> <num3> ..." << endl;
25         return 1;
26     }
27
28     for (int i = 1; i < argc; ++i) {
29         long long n = atoll(argv[i]);
30         if (n <= 0) {
31             cout << "Error: " << argv[i] << " is not a positive integer." << endl;
32             continue;
33         }
34
35         if (isGrowing(n))
36             cout << n << " is a growing number" << endl;
37         else
38             cout << n << " is not a growing number" << endl;
39     }
40
41     return 0;
42 }
43
```

```
root@DESKTOP-J52EF2R:~/projects/cpp# g++ -O2 -Wall phantranthanh Huy_itcsiu22056_l23.cpp -o grow
root@DESKTOP-J52EF2R:~/projects/cpp# ./grow 1 23 144 3689
1 is not a growing number
23 is a growing number
144 is not a growing number
3689 is a growing number
```